

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. MODULAR CODE LAYOUT:

The FitBuddy application is structured using a clean, modular layout separating the backend server routing, database ORM, external API logic, and frontend views. This architectural decision ensures the codebase remains maintainable and scalable.

- **routes.py:** Acts as the FastAPI controller. It manages the HTTP POST requests, initiates database sessions, and triggers the AI generation functions.
- **database.py:** Contains the SQLAlchemy declarative base and defines the schema for the SQLite database, explicitly mapping the User and WorkoutPlan tables.
- **gemini_generator.py:** A dedicated module configured to securely interact with the Google Gemini API (`gemini-3.5-flash-lite`), isolating the prompt engineering and exception handling from the main server logic.
- **templates/:** A directory housing the Jinja2 HTML templates used to render the user input form, the final results page, and the admin dashboard dynamically.

3. READABILITY & STANDARDS

- **Explicit Naming Conventions:** Variable and function names clearly describe their operational purpose (e.g., `generate_workout_gemini`, `WorkoutPlan`).
- **Framework Adherence:** The code utilizes FastAPI's native dependency injection (`Depends`) and form data handling (`Form( . . . )`), adhering to modern Python asynchronous web standards.

4. REUSABILITY

- **Dynamic Prompt Engineering:** The `gemini_generator.py` script is highly reusable. By utilizing f-strings (`f"Create a structured 7-day workout plan for {user_input['goal']} at {user_input['intensity']} intensity."`), the exact same function dynamically adapts to thousands of different user metric combinations without requiring code modifications.
- **ORM Portability:** By leveraging SQLAlchemy classes instead of raw SQL strings, the database schema is highly portable. The application can seamlessly transition from the lightweight SQLite database to a robust PostgreSQL server simply by altering the connection string.